

UNIVERSITY OF MINES AND TECHNOLOGY



INTRODUCTION TO DATA SCIENCE AND ANALYSIS

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PROJECT WORK REPORT

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INTRODUCTION

Data Science and Data Analysis involve extracting meaningful insights from data and using it to make decisions or using to create models that solves real world problems. One important application is **time series forecasting**, where past data is used to predict future values.

his project was to develop a predictive model capable of forecasting Tesla (TSLA) stock prices using historical market data. Stock markets are highly dynamic and influenced by many factors, making prediction is challenging but valuable task for investors and analysts. The project is based on **Time Series Forecasting**, applying both classical statistical methods and modern machine learning techniques.

PROBLEM STATEMENT

Stock market prices, especially for companies like Tesla, are highly volatile and influenced by many unpredictable factors such as market demand, news, investor behavior, and economic conditions.

The main problem in this project is how we can use historical stock data to predict future Tesla closing prices using machine learning.

Traditional forecasting methods struggle with non-linear patterns in stock data. Therefore, a machine learning approach is used to learn patterns from past price movements and predict future values.

OBJECTIVES

The objectives of this project are:

1. To analyze stock price data as a time series
2. To apply time series forecasting techniques
3. To build a predictive model using random forest (machine learning method)
4. Generate 30-day ahead forecasts

Technologies used

- Python: The main programming language used for the data analysis, model training, and forecasting.
- Pandas: The library used for data manipulation, cleaning, and time series structuring.
- NumPy: The library used for numerical computations and calculations.
- Matplotlib: The library used for visualizing the data, predictions, and forecasts.
- Scikit-learn: The library used to implement the Random Forest Regressor model.

APPROACH

1. Data Collection and Preprocessing

The dataset was downloaded from <https://finance.yahoo.com> and it consists of historical stock prices (2020-01-01 to 2026-04-09). The dataset includes the following:

- Date
- Open price
- Close price
- High and Low prices
- Volume

Steps taken to clean the data:

1. Converted the date column to datetime format and also converted all price and volume columns to float.
2. Set business day ('B') frequency to ensure continuous trading days
3. Handled missing values using forward fill
4. Ensured data is structured for time series analysis

2. Exploratory data analysis

The dataset was inspected using `.head()` and `.info()` and also, a line plot of Tesla closing price was created to observe:

- Trends
- Volatility
- Market fluctuations over time



3. Train-Test Split

The dataset was split into two and that is 80% to train our model and the remaining 20% to test it. This allows the model to learn from past data and be tested on unseen data.

4. Feature Engineering

Since stock data is time-based, the model cannot use raw values directly. In that case we had to transform our data using:

- **Lag Features:** Past values of closing price were added as inputs:
 - lag_1 (previous day)
 - lag_2
 - lag_3
 - up to lag_30
- **Rolling Features:** Moving patterns over time were used to help the model understand trends (7/14/30-day mean/std).
- **Time-Based Features:** Time-based features capture temporal patterns in the stock data that may repeat cyclically (day, month, quarter).

This converts time series data into a supervised learning dataset.

5. Model Training

We trained the Random Forest Regressor (100 trees) using the engineered features created from the dataset. These features included past closing prices (lags), which provide information about previous market behavior. The model learned the relationship between these past prices and the current closing price, allowing it to understand how earlier price movements influence the present value.

6. Model Evaluation

The model was tested on the last 20% of the dataset. Predictions were made step-by-step (rolling approach). This was to compare our model's prices to the actual prices for the remaining 20% of the dataset.

7. Forecasting Future Prices

The model was extended to predict the next 30 days. We generated the future values using previous predictions.

RESULTS

Model Prediction Performance

From the comparison plot, the model was able to follow the general trend of the stock price. Some short-term fluctuations were not perfectly captured. This is expected because random forest performs better on trend learning than sudden sharp changes.

The model was also evaluated using the following performance metrics:

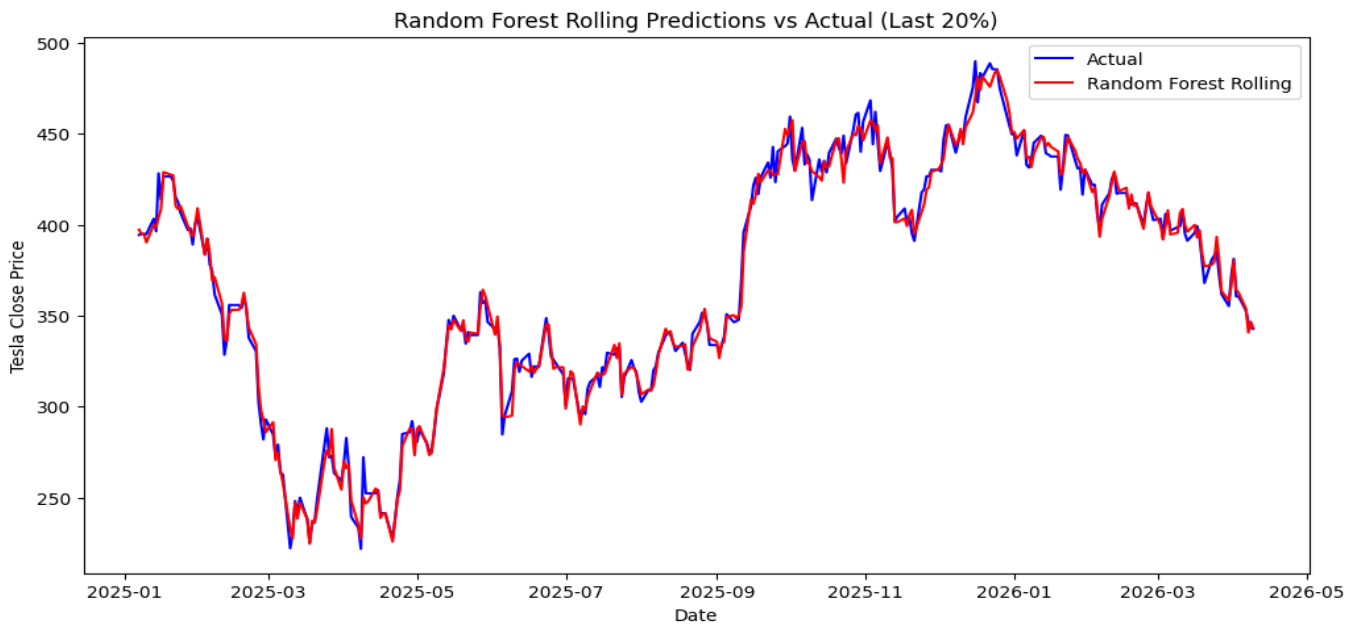
1. Mean Absolute Error (MAE)
2. Root Mean Squared Error (RMSE)

Below is a comparison metrics table:

Model	MAE	RMSE
Random Forest	4.59	6.19

Visualization

A line graph was plotted comparing the Actual stock prices and the model prediction:



30-Day Forecast prediction

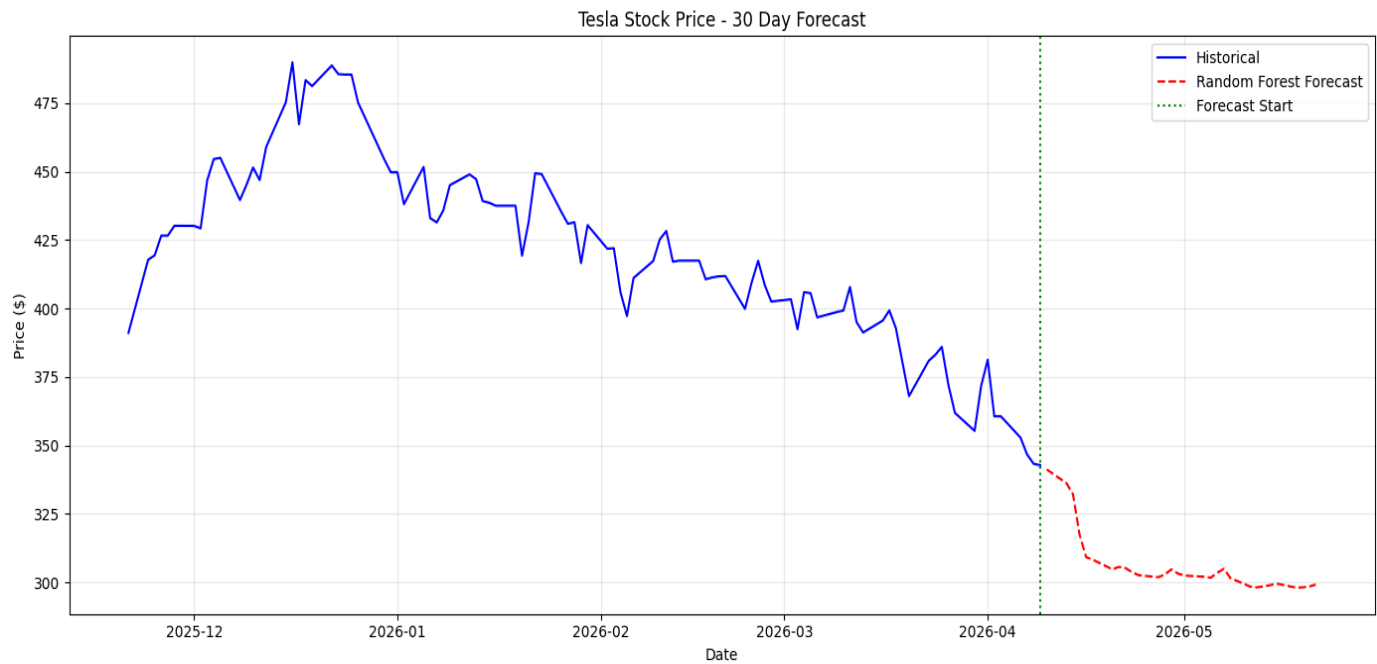
The model generated predictions for a future period spanning from April 10, 2026 to May 21, 2026, covering 30 business days.

During this forecast period, the Random Forest model predicted that the stock price would range between \$298.00 and \$341.19. This indicates that the model expects the price to fluctuate within this interval over the specified time frame.

However, it is important to note that stock prices are influenced by various external factors such as economic conditions, company performance, market news, and investor sentiment. Since these factors are not included in the model, they may affect the accuracy of the predictions and overall model performance.

A graph was plotted to show the prediction of the 30 days forecasting:

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CONCLUSION

This project successfully demonstrated how machine learning can be applied to stock market prediction.

Observations:

- Stock data must be converted into supervised learning format before modeling
- Random Forest can effectively capture trends in time series data
- Feature engineering (lags and rolling windows) is crucial for good performance
- Predictions are more reliable for trends than for exact daily values